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Cementitious Swell Packer, Development of a New Well Integrity Solution to Prevent Annulus Pressure Migration

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Abstract

The cementitious swell packer (CSP) is a product intending to replace conventional swell packers, using a swelling cement housed in a porous membrane. The CSP is prepared on a down-well pipe string section at a predetermined length, that once immersed in water, will swell over a period of days and then cure to form a concrete seal on the inner and external annulus of a pipe to pipe or pipe to rock application creating zonal isolation. This then allows a differential pressure to be applied across the CSP, currently qualified up to 2000 Psi. This method of zonal isolation has been created to replace conventional packers, allowing for a life of well seal to form that does not break down when in contact with the well interface (oil, saline water etc.) in a cost-effective manner. The CSP have the potential to have the cement formulation tailored to specific well conditions, to meet most criteria that are available for well conditions.

Introduction

The process of drilling a new typically well involves several different steps. Primarily, a bore hole is drilled to a known depth and a casing of appropriate size is inserted. Cement is then pumped through the length of the casing, out the casing shoe and into the annulus to a precalculated depth or until cement is observed at surface. A smaller bore is then drilled below the cemented casing, and the same process is repeated with a smaller diameter casing until the well has reached the area of interest. This process is typically known as well construction and the cement has two primary uses: to mechanically support the casing strings and to form a barrier between the formation sections for the well for the life of the well, this currently has an associated failure rate of anywhere between 30-50% world-wide. (Xuning, 2024)

Wells typically are made up of many different zones that are exposed during the drilling and completion process. These zones can have differing pressures within them and contain varying fluid types. For safety and environmental reasons, it is often required to isolate and contain these zones. This is typically achieved through the well construction process as previously described and through the provision of zonal isolation during the well's completion phase.

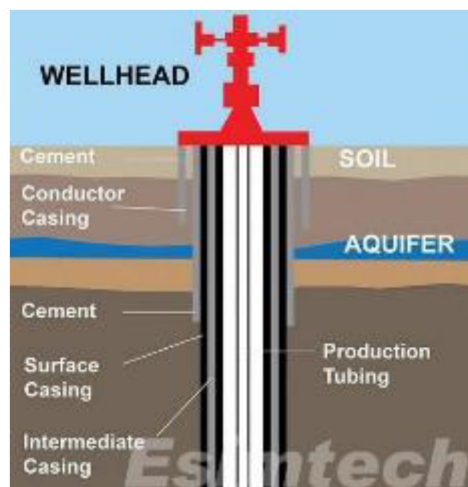


Figure 1—Typical well construction¹ (Esintech)

Cement is among the most common materials used in well construction and completions and is often the primary method employed due to its low cost, easy availability and extensive historical use in the drilling process. Despite the advances in cementing technology and operational best practices, a recurring challenge in late-life wells is the development of sustained annular pressure (SAP)—the unintended buildup of pressure in one or more annuli that persists even after bleed-off. SAP can affect 30–50% of wells worldwide (Xuning, 2024) and since 2009 it has cost the industry more than \$75 Billion² (Hart Energy, 2021). SAP can stem from various factors, including poor cement bonding, micro-annuli formation, gas migration during cement hydration, thermal cycling, and mechanical degradation of the cement sheath over time. If not adequately diagnosed and remediated, SAP poses serious risks to well integrity, environmental safety, and regulatory compliance, particularly in mature fields and aging offshore infrastructure.

Many forms of zonal isolation have been tried in the past, the most common among them are swellable elastomer packers, but they typically are not as directly compatible with cement and often exhibit failures with time. (Senters, 2016) these failure can be due to mechanical effects, temperature changes, cyclically induced mechanical loading or chemical degradation. The CSP aims to solve this issue by adding a swellable cement-based packer onto casing in the well, which is activated by well fluid to swell and form a seal, preventing and mitigating any annular leakage due to micro-annulus or cement failure. The CSP can also be classed as a primary barrier, so can be used to reduce the amount of cement required significantly on the casing string, with the potential of removing it entirely. The packer is comprised of a dry cement mix; this mix is hydrated by the well fluids with the material housing of the CSP preventing overhydration of the cement. This allows it to maintain the integrity of the mix as it is deployed into the well. The cement can also anchor the casing in place and form both mechanical support to the casing or act as more traditional packer slips and fix the casing in place in decreet locations. This could allow for a reduction in the total volume of pumped cement. The swelling profile of the cement is tuned to provide up to at least 3 days to before reaching the host casing ID and hardening. This allows time for the casing to be positioned correctly prior to use. This also ensures that a traditional cement job can be placed before the packer has expanded to its full size and fully set, so is easy to integrate into existing well programs.

Scope

The core technology for cementitious swell packers (CSP) can be applied to both cement completion and zonal isolation. The main advantages of CSPs over a traditional cement completion is the elimination of annular seal failure, and the potential to replace thousands of tonnes of pumped cement, greatly reducing running time, cost and the associated CO₂ emissions. The CSP can also replace elastomeric swell packers

with a cement-based solution that is a well-known and trusted industry standard that is considered a life of well application once applied, does not degrade in the production atmosphere and does not have the associated failure rate (approx. 30%) when compared with elastomeric packers.

The existing technology for swell packers has two major flaws. Elastomeric swell packers have accepted failure rates and relatively low operating times in comparison to life of well, or mechanical packers have high cost with significant planning and difficulty in some applications. A new technology utilizing a specially developed swelling concrete could be used to create life of well zonal isolation in a cost effective and reliable manner.

Target Testing Parameters

The development of the CSP was delivered through several phases with different testing parameters and goals.

Phase 1, Targeted the proof of concept, could a concrete be made to reliably swell to a target volume.

Phase 2 Investigated the material properties, namely, compatibility with well fluids, how to effectively pack, convey and house the swelling concrete powder, and to develop the CSP structure ensuring a reliable packer model.

Phase 3 Moved onto the prototype trials, identifying a theoretical target candidate packer size and testing the CSPs ability to withstand the required differential pressure and temperature of the identified target area.

Methods, Procedures, Process

Phase 1: Proof of Concept

Using Ordinary Portland Cement as a base, an investigation was conducted into additives that may allow for expansion of the cement, with the intention of maintaining structural integrity once cured. The target was to develop a series of concrete blends capable of expanding up to 200 % of their original volume. The testing was conducted by identifying additives that, in varying proportions, could potentially control the expansion of the concrete over a period of hours. The test involved forming a cake of concrete in a cup vessel at a water/powder ratio of 0.3, and after 24 hours, an excess of water was added on top of this, with any expansion and structural integrity observations recorded. This was tested through a simple cup test, where the concrete cake is further hydrated with an excess of water for a variety of concrete blends, shown in [figure 2](#).

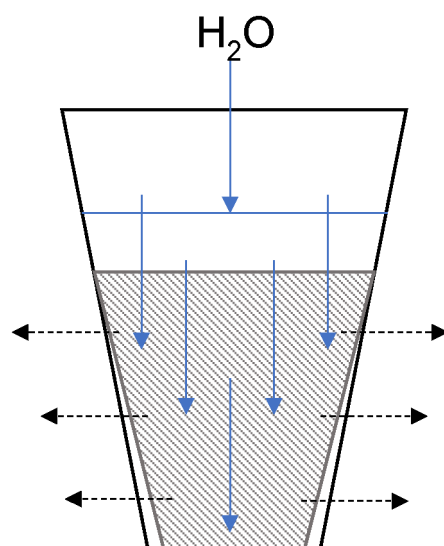


Figure 2—Cup test of concrete blend observing expansion with varying water addition

Phase 2: Material properties investigation

Testing of compatibility with saline well fluid was conducted in the same method as phase 1, shown in the [Figure 2](#) cup test, using an increasing amount of salt (NaCl) in solution up to saturated solution and observing the effect of the saline solution on the concrete blend's structural integrity by measuring the compressive strength of the expanded concrete.

The testing of conveying and housing the powder blend to achieve an acceptable density was conducted using powder conveyance technologies, and the material housing was developed in two formats. A durable elastic-based material housing, to reinforce and anchor the packer, and a friable material that decomposes under well conditions, releasing the concrete powder. These materials were designed as a toroidal shape, as discrete packets of CSP powder blends. The calculated anchoring value of the packer to withstand the differential pressures was set to a maximum of 3 foot in length and could be increased if higher pressures would be required. These factors will not be detailed in this paper as they are confidential. The description, however, will allow for the process to be understood in full.

Concept designs of a CSP model were designed and trialed using various set ups, testing the different fabric housings, on a 3-foot length of CSP, to ensure anchoring potential to determine the best practice for a CSP set up. These included the use of durable CSP toroids alone, and a combination of durable and friable CSP toroids.

Phase 3: Prototype trials

Prototypes to test the pressure differential capability of the CSPs were developed using different CSP profiles. This required the development of a pressure vessel capable of testing a full-scale prototype. A pressure rig was designed using down-well casing of appropriate size (18 5/8") to test the 13 3/8" specification CSP, using two machine caps, and M60 rods, tensioned to allow for a seal up to 3000 Psi, shown in [Figure 3](#). A series of ceramic heater bands were attached to the casing to allow temperature control, with the whole vessel then housed in an insulated box, to ensure temperature regulation.



Figure 3—Pressure Vessel Test Rig

A control panel, shown in [Figure 4](#), was set up using Swagelok fittings, to allow accurate control and measurement (using pressure transducers) of the pressure input and output of the fluid in the casing, with a hydraulic pump used to alter pressure within the vessel. This was recorded on a program, measuring in situ pressure changes during the testing protocol.



Figure 4—Pressure Control Panel

Results & Observations

Phase 1

The most promising blends were identified after the addition of the excess of water, with expansion and coherence being the two factors measured. An excess in expansion was related to over hydration of the sample, forming soft, non-coherent samples. Flash setting occurred with some samples, in which the samples were brittle and cracked with further addition of water. The samples that were most promising, showed an increase in volume, but maintained structural integrity. The mechanical compressive strength was measured of these samples after a period of 7 days under the specified conditions obtained from field conditions. The conditions for testing are shown in [table 1](#).

Table 1—Conditions set for testing of CSP material, specifying a 13 3/8" packer

Application Name	Packer Base pipe OD / Inches	Base Pipe Grade / lbs/ft	Packer Base Pipe ID / Pass Through	Packer Running OD / Inches	Casing Setting ID / Inches	Differential Pressure / Psi	Element Length / Inches	Retard Time / Days	Swelling Time / Days
13-3/8" × 18-5/8" Swell Packer	13.375	72	12.559	15.25	17.559	2000	36	2	5

The compressive strength of the four most promising samples, set at 55 °C and 2000 Psi is shown in [table 2](#) below.

It was found that the most promising blend gained strength in the first 7 days, with a delay in mechanical strength gain until day 4. Consecutive packers can be stacked to provide a linear increase in anchoring capability, so the cumulative strength allows for pressure resistance of up to 2500 Psi (at 13 3/8 Size), with 3 feet of CSP in place. This value can be tuned to increased anchoring potential if required by increasing the length of the CSP. The nature of the blend design allows for control over the strength gain of the concrete, enabling tuning for specific well conditions.

Table 2—Compressive strength at time / days

CSP Blend Type	Compressive Strength / Psi		
	7 days	14 days	28 days
TWD3.2	2465.6	4641.2	6526.7
TWS3.1	1595.4	3625.9	4641.2
TOD4	1740.5	4061.1	5366.4
TS1.5	2030.5	4206.1	5801.5

The successful development of the concrete blend had shown that the key to the proof-of-concept testing was identified to be not exceeding the critical water/powder ratio required, with TWD3.2 exhibiting a concrete that expanded by 180% of its original volume and of sufficient strength to withstand the forces that would be seen under a differential pressure of 2000 Psi.

The water/powder ratio requirement informed the specific packing density required to control the water intake into the toroidal material.

Phase 2 & Phase 3

Initial testing of the TWD3.2 blend showed a promising interaction with saline fluid. The data in [table 3](#), shows that the compressive strength is unaffected by the salt solution and that there is a positive impact of increasing the brine to a higher % of salt inclusion. Whilst this result was somewhat unexpected, salt can lead to numerous problems with concrete, in particular increasing localised hydration, which can lead to a higher water/powder ratio than desired. Some of the additives in the concrete blend design do suggest there could be synergistic effects with the components allowing for hydration of the expansive phases to stabilise further by drawing more water to the expansive phase during the further curing of the concrete, leading to higher compressive strengths in the latter stages of the concrete reaction.

Table 3—Compressive strength a time as salt (NaCl) content of water is increased / days

Salt (NaCl) content of brine / %	Compressive Strength / Psi		
	7 days	14 days	28 days
10	2320.6	4351.1	6526.7
15	2465.6	4496.2	6526.7
25	2465.6	4641.2	6526.7
36	2610.7	5076.3	6816.8

This result shows that hydration with completion fluid is possible, up to and including saturation of salt (NaCl) in the solution. The advantage of this finding is that the CSP can be hydrated once in place by the completion fluid, triggering the curing process of the CSP.

Filling the durable material housing is a complex process, that requires active fluidisation of the powder, to allow for a high density fill in the material. The process is completed by densifying the powder in the toroidal material using a specialised compaction and distribution jig, shown in [figure 5](#). This method allows for the concrete powder to be highly compacted, preventing overhydration of the concrete, and forming a regular square ended, discrete toroid. The durable toroids were the first design in the CSP process, with the idea that each toroid would be able to form individual discrete seals, with the cumulative compressive strength of the concrete able to withstand the differential pressure force. Several pressure tests were run with durable toroids alone, with a completed packer shown in [figure 6](#).

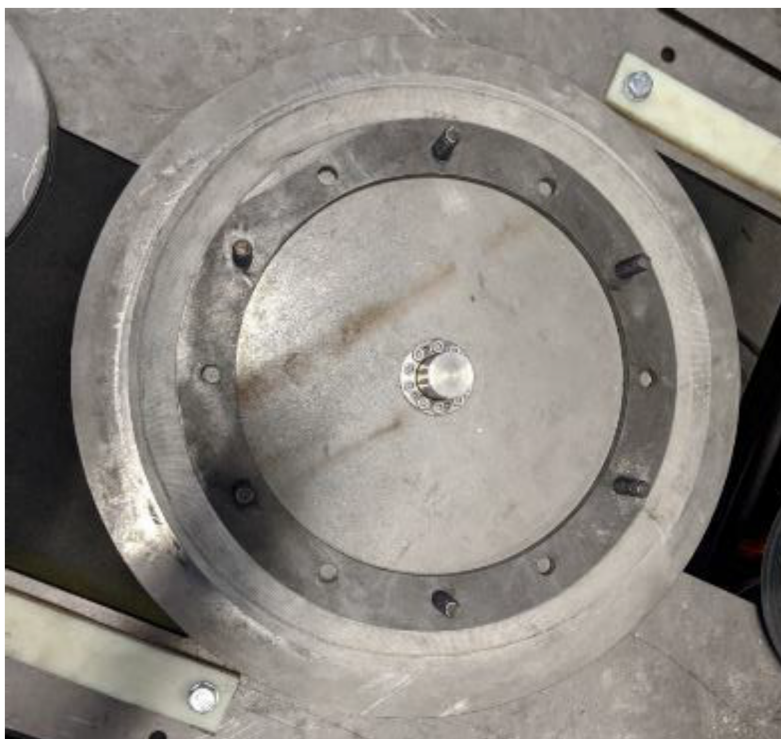


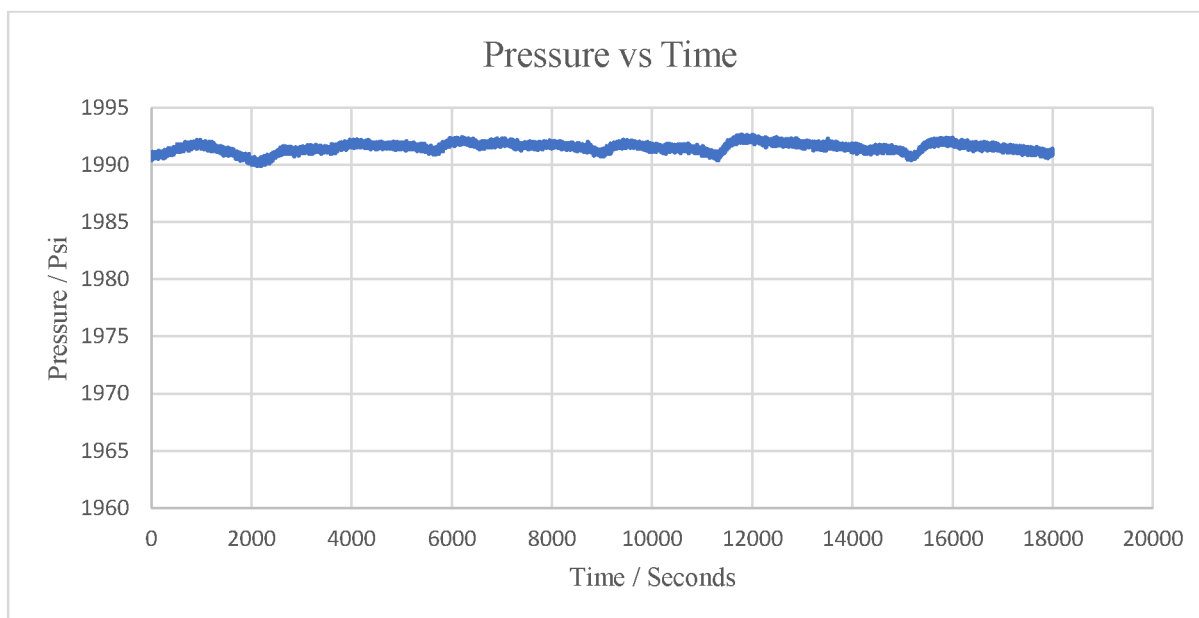
Figure 5—Compaction and distribution jig for durable CSP.



Figure 6—Durable toroid only CSP.

Whilst the test was successful in proving the full-scale concept, there were difficulties with maintaining pressure and temperature accurately throughout the test process leading to significant difficulty in analysing

the results. What was found after successive testing, however, was that the prototypes were able to act as significant fluidic resistors, but not capable of holding a complete seal, as can be seen from the pressure data over 5 days shown in [Graph 1](#). Further analysis shows this slow leak rate trends downwards over time.



Graph 1—Pressure vs Time for durable only CSP

When the casing was cut open to attempt to identify the cause of the leak through the CSP, it was found that a pattern from the material was left on the annulus and the external casing faces, which can be seen in [Figures 7 & 8](#).



Figure 7—Exhumed Pressure test of durable CSP.



Figure 8—Material pattern artifact on inner casing annulus.

Whilst the pattern is difficult to observe on the external casing, the pattern is extremely clear on the inner annulus. There is also evidence of rust forming on the inner annulus, likely exacerbated from the contact with salt water. The regular pattern from the material is also visible on the concrete itself, providing a plausible leak path for the fluid to take. This led to further innovation in material housing, and the development of a friable toroid, capable of decomposing when in contact with the well fluid. The design of the friable toroid was to include no inner annulus interface with the material, enabling direct contact of the concrete powder with the casing, leading to a higher likelihood of a seal between the casing and concrete. This was made possible by alternating the durable toroid with the friable toroids, using the durable toroids to form a backup and anchoring seal, with the friable sections acting as the pressure seal. An excavated version of this design can be seen in Figure 9, where the concrete only sections can be seen clearly next to the material housed in the durable toroid sections. This packer was tested under the same conditions of 55 °C, and a 2000 Psi differential and provided a stable pressure trace at 2000 Psi over a 30-day trial, proving that the CSP works for the intended purpose. The next stage of the project is the development of a CSP manufacturing facility in the Kingdom and conducting a down-well trial for qualification.

Durable material

Friable material



Figure 9—Durable and Friable CSP prototype

Conclusions

Based on the operations conducted and explained in this paper, the following can be concluded:

- A concrete blend containing OPC as a base can be used to form a swelling cement capable of achieving a high compressive strength with good coherence after 7 days at 17 MPa and after 28 days at 45 MPa.
- When a salt (NaCl) solution is used to hydrate the CSP concrete blend, the compressive strength is slightly elevated when compared to using water alone.
- A full-scale pressure testing apparatus was successfully developed to test the CSP prototypes in house.
- A specially designed housing material can be produced to encapsulate the concrete component and control the hydration during the curing process, preventing the concrete from being washed away or over-hydrated.
- Durable CSP toroids alone act as fluidic resistors, but due to the artifact pattern of the material left during the curing process, a full seal is not possible with the durable section alone.
- The use of a dissolvable, or friable, material housing allows for a direct cement seal, preventing leakage through the inner annulus.
- A 3-foot packer comprised of multiple sections of durable and friable toroid sections containing the CSP powder can hold a differential pressure of 2000 Psi over a period of 30 days at 55 °C.

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Abbreviations

- CSP – Cementitious swell packer
- TW3.2 – Concrete blend with ternal white inclusion, third test bracket, second proportion selection.
- TW3.1 – Concrete blend with ternal white inclusion, third test bracket, first proportion selection.
- TOD4 – Concrete blend with ternal white inclusion, organic inclusion, fourth test bracket, base blend.
- TOS1.5 – Concrete blend with ternal white inclusion, organic inclusion, silica-based aggregate, first test bracket, fifth proportion selection.
- Durable toroid – Material housing for CSP that does not break down in well atmosphere.
- Friable toroid – Material housing for CSP that degrades and dissolves in well atmosphere.
- Toroid – A ring shaped object that is generated by rotating a closed curve around an axis that doesn't intersect the curve.

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